

Introduction to Programming

Lecture 6 — Objects in your programs

Nicolas Gry

Université de Saint-Étienne | Year 2026/2027

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-1 Recap'

00 Assignment review

01

Defining your own types

An example of `struct`

Let's walk through an example.

└ An example of `struct`

Explain that `std::array<std::array<std::array<int, cols>, rows>, 3>` does not encapsulate the color channel order. Hence the need for a `Pixel` structure.

Show instantiation + assigning value to member variable.

Custom types and member variables



You can create a type with `struct`.

A type can contain **member variables**. When you instantiate a variable of a custom type, you are implicitly instantiating all those member variables.

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```
// Definition -> no instance in memory yet
struct Pixel {
    int r,g,b;
}; // Don't forget the semicolon here !

// Instanciation, p exists in memory
Pixel p;
p.r = 42;
```

02

Member functions

Another cool example

You know the drill: <https://godbolt.org/z/8Khhqocc3>.

└ Another cool example

Implement `height()` and `width()`. Show them you can put the definition outside of the structure body. Show them how to call the function.

Declaring and defining member functions



A custom type can have **member functions**. You declare them in your structure body; you can put the definition either in the structure body or outside of it. In the latter case, the function's name has to be preceded by a double colon “:”.

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```
struct Foo{  
    void a_function() {}  
};  
  
struct Foo{  
    void a_function();  
};  
void Foo::a_function() {}
```

```
struct Foo{  
    void a_function() {}  
};  
  
struct Foo{  
    void a_function();  
};  
void Foo::a_function() {}
```

Calling a member function

Once you have an instance of a custom type, you can call the member function with that instance.

└ Calling a member function

At the end of this slide, go back to your example code on Compiler Explorer. Try to implement a `info(const Image&)` function that displays the dimension. It will fail because `width` and `height` are not `const`. Introduce `const` functions.

Calling a member function

Once you have an instance of a custom type, you can call the member function with that instance.

```
struct Number {  
    int val;  
    void increment() { val++; }  
};  
Number a {1};  
Number b {10};  
a.increment(); // a.val == 2  
b.increment(); // b.val == 10
```

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Const functions 1/2

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```
struct Foo{  
    void func() const {} // func is const  
};
```

Const functions 2/2

When passing an object as a `const` argument to a function, only `const` member functions can be called in that function

Const functions 2/2

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```
struct Song{
    std::string name;
    std::string name () const { return name; }
    void rename(const std::string new_name) { name = new_name; }
};

void display(const Song& song){
    std::cout << song.name() << std::endl; // ok
    song.rename("hello"); // not ok, rename is not const
}
```

```
struct Song{
    std::string name;
    std::string name () const { return name; }
    void rename(const std::string new_name) { name = new_name; }
};

void display(const Song& song){
    std::cout << song.name() << std::endl; // ok
    song.rename("hello"); // not ok, rename is not const
}
```

Overloads

Just like *free* functions — regular functions —, you can *overload* member functions following the same rules as before.

```
struct RGBImage {  
    std::vector<std::vector<Pixel>> pixels;  
  
    void set_pixel(int x, int y, int r, int g, int b);  
    void set_pixel(int x, int y, Pixel pixel);  
};
```

03

Public vs Private members

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Abstraction needs

You may want to hide some information contained in your type. Things that should be available only inside your structure, but not outside of it should be **private**. Things available to everyone will be **public**.

```
struct MyType{
public:
    int everyone_can_see_me;
    int compute() {
        return everyone_can_see_me
            + private_part;
    }
private:
    int private_part;
};
```

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    int compute() {
        return everyone_can_see_me
            + private_part;
    }
private:
    int private_part;
};
```

```
MyType var;
var.compute(); // ok
var.private_part = 7; // not ok,
                    won't compile
```

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class **VS** struct

Another keyword to define a type is `class`.

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Members of a `struct` are `public` by default (`strUct` → `pUbl`ic).

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class **VS** struct

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Members of a `struct` are `public` by default (`strUct` → `pUbl`ic).

Members of a `class` are `private` by default (`clAss` → `privAte`).

```
struct Foo{  
    int var; // public  
};
```

```
class Foo{  
    int var; // private  
};
```


04

Some special functions: constructors and destructors

Constructors through an example



Let's dive in here: <https://godbolt.org/z/TbPvevjfc>

Constructors 1/2

A **constructor** is a function that is called whenever a variable is instantiated.
The constructor that takes no argument is called the *default* constructor.

```
struct Pixel {  
    int red, green, blue;  
};  
Pixel p1; // default constructor called
```

Constructors 2/2

You can overload the constructor with one that takes argument. If you still want the default constructor to be available, you have to define it.

Constructors 2/2

You can overload the constructor with one that takes argument. If you still want the default constructor to be available, you have to define it.

```
struct Pixel {  
    int red, green, blue;  
    Pixel () = default;  
    Pixel(int r, int g, int b)  
        : red{r}, green{g}, blue {b} {}  
};  
Pixel p1; // default constructor called  
Pixel p2 {111, 22, 3}; // custom constructor
```

```
struct Pixel {  
    int red, green, blue;  
    Pixel () = default;  
    Pixel(int r, int g, int b)  
        : red{r}, green{g}, blue {b} {}  
};  
Pixel p1; // default constructor called  
Pixel p2 {111, 22, 3}; // custom constructor
```

Destructors 1/2

Conversly, a **destructor** is a function that is called whenever a variable is *destroyed* (i.e it goes out of scope).

The destructor never takes any argument.

Destructors 2/2

```
struct AudioFile {
    AudioFile(const std::string filename) {
        m_file = sf_open(filename.c_str(), SF_READ, &m_info);
    }
    ~AudioFile(){
        sf_close(m_file);
    }
private:
    SF_INFO m_info;
    SNDFILE * m_file;
};
```

```
struct AudioFile {
    AudioFile(const std::string filename) {
        m_file = sf_open(filename.c_str(), SF_READ, &m_info);
    }
    ~AudioFile(){
        sf_close(m_file);
    }
private:
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};
```

05

Abstraction

Why do we need our own types?



It's a way to group things that go together.

It's easier to manipulate variables and functions that all relate to the same thing when they're grouped in the same place.

How do we design types?

Think about how the object will be used. Think in terms of functionality.
The *public interface* should expose *what you can do* with an object; *private members* contain *how the object implements it*.
This is called **abstraction**.

Abstraction example

Without abstraction

```
std::vector<std::vector<int>>>
    bw_image = {
        {12, 125}, {45, 242}};
int height = bw_image.size();
int width = bw_image[0].size();
bw_image[0][1] = 43;
```

With abstraction

```
BWImage image;
image.height();
image.width();
image.set_pixel(0, 1, 43);
```

Abstraction example

Without abstraction

```
std::vector<std::vector<int>>>
    bw_image = {
        {12, 125}, {45, 242}};
int height = bw_image.size();
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```

With abstraction

```
BWImage image;
image.height();
image.width();
image.set_pixel(0, 1, 43);
```

One class = One responsibility



You don't want to overcrowd a class code. Think in terms of what your class is responsible for, what it should represent.

In the case of an image, you split the definition of a pixel and the definition of an image.

One class = One responsibility

You don't want to overcrowd a class code. Think in terms of what your class is responsible for, what it should represent.

In the case of an image, you split the definition of a pixel and the definition of an image.

This makes your code more *maintainable*. The more modular your code, the easier it is to modify a piece without breaking every other ones.

Member function vs Free function

Member functions

Operate on private members / internals.

Free function

Whenever possible, make a function a free function. It could be used similarly on other types.

Practise time !

06

An introduction to generic programming

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Making a function work on many types

An example.

└─ Making a function work on many types

Rewrite `print_dim` as a generic function. Show them a more complex example with brightness function and intuit that you want to operate at the image level => you write an overload for the pixel brightness and a generic function for the image brightness.

Generic functions

A **generic function** is a function that can operate on values of different types within a single definition.

```
template<typename ImageType>
void print_dimensions(const ImageType& image) {
    std::cout << "width: " << image.width()
               << std::endl;
    std::cout << "height: " << image.height()
               << std::endl;
}
```

```
template<typename ImageType>
void print_dimensions(const ImageType& image) {
    std::cout << "width: " << image.width()
               << std::endl;
    std::cout << "height: " << image.height()
               << std::endl;
}
```

Type deduction 1/2

Type deduction happens the parameter of a generic function can be inferred by the call of the function. Type is usually deduced from passed arguments' type.

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Type deduction happens the parameter of a generic function can be inferred by the call of the function. Type is usually deduced from passed arguments' type.

```
template<typename T>
T add(T a, T b) { return a+b; }
add(1, 2); // T is int
add("hello ", "world"); // T is char *
```

Type deduction 2/2

A function parameter cannot be deduced from the return type alone. When type deduction is not possible, you have to specify the parameter's type yourself.
Example.

Type deduction 2/2

A function parameter cannot be deduced from the return type alone. When type deduction is not possible, you have to specify the parameter's type yourself.

Example.

```
template<typename Image>
Image create_image(int w, int h){
    return Image(w,h);
}

RGBImage image = create<RGBImage>(50, 50);
```

Generic functions and multiple files



You **cannot** forward declare a generic function.

If you want to share a generic function using different files, its definition will have to go into a header file.

Practise time!

See you next time!